



Low-Cost AIR-QUALITY METER

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Air pollution and concern about air quality is not something new. Complaints were recorded even in the 13th century when coal was first used for industrial purposes in London. From the middle of the 19th century, the atmosphere of major British cities was regularly polluted by coal smoke in winters, giving rise to the infamous mixture of fog and smoke known as smog. Today the emphasis has shifted from pollution problems caused by the industry

to the ones associated with motor vehicle emissions.

Air Quality Index

Air Quality Index, or AQI, is a number used by government agencies to communicate to the public how polluted the air currently is, or how polluted it is likely to become. As AQI increases, an increasingly large percentage of the population experiences severe adverse health effects.

Different countries have their own air-quality indices correspond-

ing to different national air-quality standards. Some of these are Air Quality Health Index (Canada), Air Pollution Index (Malaysia) and Pollutant Standards Index (Singapore).

The national AQI was launched in New Delhi on September 17, 2014, under Swachh Bharat Abhiyan. There are six AQI categories (Table I) of the same. The proposed AQI considers eight pollutants (PM10, PM2.5, NO₂, SO₂, CO, O₃, NH₃ and Pb) for which short-term (up to 24-hour averaging period) National Ambient Air Quality Standards are prescribed.

For details, you may refer below link

<http://pib.nic.in/newsite/PrintRelease.aspx?relid=110654>

In this project the sensors used are SDS011, MQ-135, DHT11 and DS1307 (as shown in Fig. 1).

A typical PM2.5/10 based meter costs around US\$ 200. Here we present a meter for monitoring air particles, gas concentration, temperature, humidity and time, which costs around US\$ 45 only. Block diagram of the air-quality meter is shown in Fig. 1.

Circuit and working

Circuit diagram of the air-quality meter is shown in Fig. 2. Heart of

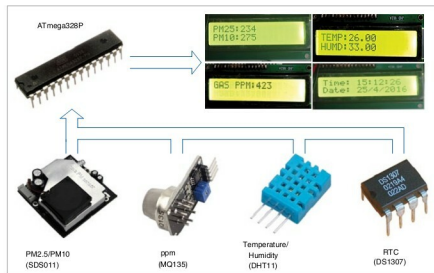


Fig. 1: Block diagram of the air-quality meter

TABLE I
AQI CATEGORIES

AQI Category (Range)	PM10 (24hr)	PM2.5 (24hr)	NO ₂ (24hr)	O ₃ (8hr)	CO (8hr)	SO ₂ (24hr)	NH ₃ (24hr)	Pb (24hr)
Good (0-50)	0-50	0-30	0-40	0-50	0-1.0	0-40	0-200	0-0.5
Satisfactory (51-100)	51-100	31-60	41-80	51-100	1.1-2.0	41-80	201-400	0.6-1.0
Moderately polluted (101-200)	101-250	61-90	81-180	101-168	2.1-10	81-380	401-800	1.1-2.0
Poor (201-300)	251-350	91-120	181-280	169-208	11-17	381-800	801-1200	2.1-3.0
Very poor (301-400)	351-430	121-250	281-400	209-748	18-34	801-1600	1201-1800	3.1-3.5
Severe (401-500)	430+	250+	400+	748+	34+	1600+	1800+	3.5+